

Python Programming Guide

Introduction to Python

Python is a high-level, interpreted programming language known for its simplicity and readability. It was created by Guido van Rossum and first released in 1991. Python supports multiple programming paradigms including procedural, object-oriented, and functional programming.

Data Types

Python has several built-in data types including integers, floats, strings, lists, tuples, dictionaries, and sets. Variables are dynamically typed, meaning you do not need to declare the type explicitly before using them.

Functions and Classes

Functions in Python are defined using the `def` keyword. Classes support object-oriented programming with attributes and methods. Python supports inheritance, encapsulation, and polymorphism as core OOP principles.

Popular Libraries

Python has a rich ecosystem of libraries. NumPy and Pandas are used for data manipulation. Matplotlib and Seaborn for visualization. Scikit-learn for machine learning. TensorFlow and PyTorch for deep learning. FastAPI and Flask for web development.